ECON3700
ENVIRONMENTAL ECONOMICS

Final Review

Guidelines to preparing an applied case study

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What is environmental economics?



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REUTERS/Carlos Barria 2011



What is environmental economics?

Questions environmental economists try to answer:

1. What is the right trade-off between **development** / **pollution** / **environmental quality**?

2. Is society on track to achieve it?

3. If not:
   i. Why?
   ii. How might we get there?

Some answers from environmental economics

1. What is the right trade-off between **development** / **pollution** / **environmental quality**?
   - Tools: theory
   - Answers: MAC=MD

Some answers from environmental economics

2. Is society on track to achieve it?
   - Tools: CBA, revealed and stated preference analysis, integrated assessment models, econometric analysis
   - Answers:
     - Major policy achievements in some domains e.g., air quality
     - Environmental improvements could have been achieved at lower cost
     - Major challenges/inefficiencies remain e.g., climate change

Some answers from environmental economics

3. If not, why?
    - Market failures like public good problems are hardest to solve:
    - With global pollution problems
    - When abatement is costly
    - When benefits accrue far into the future
    - When damages are not known with certainty

Some answers from environmental economics

3. If not, how might we get there?
   - Ideally by pricing externalities
   - If not, at least by picking cost-effective policies that:
     - Have positive net benefits (benefits – costs)
     - Have the highest BCR
     - Account for heterogeneity in MAC

Topics in this course

2. Evaluating solutions
2.1 Decentralised policies
2.2 Centralised policies
2.3 Market-based policies

Introduction | Analytical Tools | Environmental Analysis | Environmental Policy Analysis | **Environmental Issues**

1. Understanding the problem
1.1 Market allocation and market failures
1.2 Optimal pollution/emissions
1.3 Cost benefit and cost-effectiveness analysis

3. Applications


Global climate change



Globalization – trade & the environment


Economic development & the environment


Air pollution

Public policy approaches / Criteria for evaluating policy analysis

Decentralized policies

- Liability laws
- Property rights
- Voluntary action

Centralized policies

- Command-and-control
- Standards

Market-based policies

- Taxes and subsidies
- Transferable discharge permits (cap-and-trade)

- Efficiency
- Cost-effectiveness
- Fairness
- Enforceability
- Flexibility
- Incentives for technological innovations
- Moral considerations

APPLIED CASE STUDY OVERVIEW

Part 1 Written Report MUST be submitted as a Word file (.doc or docx)

Part 1 Spreadsheet Answers MUST be submitted as an Excel file (.xls or .xlsx)

Part 2 Reflective Writing MUST be submitted as a Word file (.doc or .docx)

***Part 1**: 100 marks (30%) and **Part 2**: 10 marks (5%)*

Refer to the task sheet for more details.

How do I write a 1500-word industrial report?

1. Title
2. ***Executive Summary***
3. ***Introduction***
4. ***Methodology***
5. ***Key Assumptions and Data***
6. ***Results and Analysis***
7. ***Recommendations***
8. ***Conclusion***
9. References

How do I write a 1500-word industrial report?

1. Title

Word count excluded.

Written Report on Ecological Health and Human Impacts on the Coorong South Lagoon.

2. Executive Summary

Briefly state the objective of the report. Summarize key results. Include a recommendation.

3. Introduction

Refer to the task sheet and Guest Lecture recording.

How do I write a 1500-word industrial report?

4. Methodology

CEA and CBA.

5. Key Assumptions and Data

Outline data sources.

Acknowledge limitations or uncertainties.

6. Results and Analysis

CEA for each options – Table.

CBA for each options – Table.

Comparison of CBA and CEA results.

Include graphs if you think they are appropriate or relevant.

How do I write a 1500-word industrial report?

7. Recommendations

Provide a recommendation on the most economically feasible restoration option.

8. Conclusion

Summarise the key findings and reiterate its importance.

9. References

Include a list of all the references used in your research brief and infographic. APA 7th

REFLECTION – 10 marks (5%)

In your reflective writing, use the **5R's framework** (Reporting, Responding, Relating, Reasoning, and Reconstructing) to address the following prompt:

Reflect on your experience with the Industry Expert during their presentation. What insights did you gain about the application of CEA and CBA in real-world environmental projects? How did this experience shape your understanding of the challenges and opportunities involved in addressing ecological decline and balancing environmental, social and economic outcomes?

Strict 400-word limit

REFLECTION – 10 marks (5%)

Reporting	Report what happened or what the issue or incident involved and why it is relevant.
Responding	Respond to the incident /issue by making observations, expressing your opinion, or asking questions.
Relating	Relate or make a connection between the incident or issue and your own skills, professional experience, or discipline knowledge. Have I seen/experienced this before? Do I have the skills/knowledge to deal with this?
Reasoning	Highlight significant factors underlying the incident or issue. Explain their significance for understanding the incident or issue. Consider different perspectives.
Reconstructing	Reframe or reconstruct for the future. How would I deal with this next time? What might work and why? Are there different options? What have I read/learned in the course which might be relevant?

Reference: Bain, J. D., Mills, C., Ballantyne, R., & Packer, J. (2002). Developing Reflection on Practice Through Journal Writing: Impacts of variations in the focus and level of feedback. *Teachers and Teaching*, *8*(2), 171–196. https://doi.org/10.1080/13540600220127368

REFLECTION – 10 marks (5%)

In general,

(1) What were your **assumptions** before the Industry Expert presentation?

(2) How have your assumptions changed (after the Industry Expert presentation and doing this task)?

Examples, not limited to these points.

Reporting

What was the primary focus of the case study, and what role did your industry partner play in it?
Summarize the key takeaways from the industry expert's presentation.

Responding

How did working on a real-world problem impact your learning experience?
What emotions did you experience (e.g., frustration, excitement) while analyzing the case study with real-world data?

Relating

How does this case study connect to environmental economics theories or concepts covered in class?
Can you relate this experience to future career goals or other real-world applications?

Reasoning

How did the collaboration with the industry partner influence the way you approached the analysis?
What trade-offs or policy implications did you identify in balancing environmental goals and economic outcomes?

Reconstructing

If given more time or resources, how would you approach this case study differently?
How has this experience changed your perspective on the role of economics in solving environmental challenges?

Evaluations

Important! Please complete…

https://eval.uq.edu.au/



THANK YOU!!

KEEP IN TOUCH >>>

Dr Ammarr Aripin

Research interests

- Environmental economics, natural resources economics.
- Fisheries and aquaculture economics.
- Climate change is REAL!
- Teaching and learning of CBA.

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